BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: 8191

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics - Paper: II - Objective (i)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8191

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. Lim $(\theta \to 0)\ (1-\cos\theta)/\theta$ = ...

(A) 1
(B) 0
(C) -1
(D) ∞

Q2. Lim $(n \to \infty)\ (1-1/n)^n$ = ...

(A) e^{-1}
(B) e
(C) e^2
(D) $1/e^2$

Q3. If $x = at^2$, $y = 2at$, then dy/dx = ...

(A) 2/y
(B) 2a/y
(C) 2ay
(D) 2a

Q4. If f(x) = e^(√x−1), then f'(x) = ...

(A) 1/(2√x) e^(√x−1)
(B) e^(√x−1)
(C) 1/2x
(D) e^(√x−1)/√x

Q5. If f(x) = cos x, then f'($\sin^{-1} x$) = ...

(A) −sin x
(B) −x
(C) 1
(D) $1/\sqrt{(1-x^2)}$

Q6. If y = e^(2x), then y_4 = ...

(A) 16e^(2x)
(B) 8e^(2x)
(C) 4e^(2x)
(D) −16e^(2x)

Q7. ∫ (x+2)/(x+2) dx = ...

(A) 1+c
(B) x+c
(C) −x+c
(D) 2x

Q8. The order of differential equation $y\,(dy/dx)^2 + 2x = 0$ is:

(A) [illegible in source scan]
(B) −1
(C) 2
(D) −2

Q9. ∫_{π/6}^{[illegible in source scan]} cosθ dθ = ...

(A) (√3−1)/2
(B) √3/2
(C) 1+1/√3
(D) 2/√3 − 1

Q10. ∫ ln x dx = ...

(A) $(\ln x)^2$+c
(B) x ln x + c
(C) x ln x − x + c
(D) −x ln x + c

Q11. Equation of horizontal line through (a,b) is

(A) y = a
(B) x = a
(C) x = b
(D) y = b

Q12. Two lines l_1 and l_2 with slope m_1 and m_2 are perpendicular if:

(A) $m_1 = m_2$
(B) $m_1m_2 = 1$
(C) $m_1m_2+1 = 0$
(D) $m_1 = 2/m_2$

Q13. The lines represented by $ax^2+2hxy+by^2 = 0$ are parallel if:

(A) $h^2 - ab = 0$
(B) $h^2 - ab > 0$
(C) $h^2 - ab < 0$
(D) $h^2 = a + b$

Q14. A solution of x + 2y < 6 is:

(A) (8,0)
(B) (0,8)
(C) (5,1)
(D) (1,2)

Q15. The line y = mx + c intersect the circle $x^2+y^2 = a^2$ at most _____ point/s

(A) One
(B) Two
(C) three
(D) Infinite

Q16. Equation of point circle is:

(A) $x^2+y^2 = 1$ (B) $x^2+y^2 = -1$

(C) $x^2+y^2 = 0$ (D) $x^2-y^2 = 0$

Q17. Focus of parabola $x^2 = 4ay$ is

(A) (a,0) (B) (0,a)

(C) (−0,0) (D) (0,−a)

Q18. For hyperbola value of eccentricity is:

(A) $e = 0$ (B) $e < 1$

(C) $e = 1$ (D) $e > 1$

Q19. $(i \times k)$ = ...

(A) i (B) −j

(C) 0 (D) 1

Q20. The projection of $a = i - j$ along $b = j + k$ is:

(A) $1/\sqrt{2}$ (B) $-1/\sqrt{2}$

(C) 1 (D) −2

Mathematics - Paper: II - Subjective

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: - Section B is compulsory. Attempt any three questions from section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Express the volume V of a cube as a function of the area of its base

ii. For any real valued function, $g(x) = 1/\sqrt{x^2}$; $x \neq 0$, find gog(x)

iii. Evaluate Lim $(x\to0)$ $(1-\cos x)/\sin^2 x$

iv. Differentiate $(2x-3)/(2x+1)$ w.r.t. x

v. Find dy/dx if $x^2-4xy-5y = 0$

vi. Differentiate $(\sin 2\theta - \cos 3\theta)^2$ w.r.t. θ

vii. Find f'(x) if f(x) = ln(e^x + e^(−x))

viii. Find y_2 if $y = x^2$ e^(−x)

ix. Apply Maclaurin series expansion to prove that $\sqrt{(1+x)} = 1 + x/2 - x^2/8 + x^3/16 + \ldots$

x. Find the extreme values for the function, $f(x) = x^2 - x - 2$

xi. Define objective function.

xii. Graph the solution set of the inequality, $3x - 2y \geq 6$

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Evaluate ∫ (3x+2)/√x dx, (x > 0)

ii. Evaluate ∫ (e^(2x) + e^x)/e^x dx

iii. Evaluate ∫ sec x dx

iv. Evaluate ∫ x/√(4+x²) dx

v. Evaluate ∫ a^x x dx (a > 0, a ≠ 1)

vi. Evaluate ∫ (x+b)/(x²+2bx+c)^(1/2) dx

vii. Evaluate definite integral ∫_1^2 (x²+1) dx

viii. Find the volume of the parallelepiped determined by, u = i + 2j − k, v = i − 2j + 3k, w = i − [illegible in source scan]

ix. Find the value of [i i k]

x. Find the constant α such that the given vectors are coplanar. i − j + k, −i − 2j − 3k and 3i − α [illegible in source scan]

xi. Write the formula to find the volume of tetrahedron.

xii. Write two properties of cross product.

Q4.. (Marks 18) Write short answers to any Nine parts. (9 x 2 = 18)

i. Find distance between A and B, midpoint of AB, where A(3,1), B(−2,−4)

ii. Find slope and inclination of line joining A(3,−2) and B(2,7)

iii. By means of slope, show that (−1,−3),(1,5) and (2,9) are collinear.

iv. Find equation of line through A(−6,5) having slope 7

v. Find point of intersection of x−2y+1 = 0 and 2x−y+2 = 0

vi. Find the lines represented by $2x^2+3xy-5y^2 = 0$

vii. Find the distance from P(6,−1) to the line 6x−4y+9 = 0

viii. Find the equation of circle with centre (5,−2) and radius 4

ix. Write down equation of tangent and normal to $x^2 + y^2 = 25$ at (4,3)

x. Write equation of parabola with Focus (1,2); vertex (3,2)

xi. Find foci and eccentricity of the ellipse $x^2 + 4y^2 = 16$

xii. Find equation of hyperbola with centre (0,0) Focus (6,0) and vertex (4,0)

xiii. Find centre and radius of circle $4x^2 + 4y^2 - 8x + 12y - 25 = 0$

> Note: SECTION-C (Note: Attempt any Three questions. Each question carries 10 marks)

Q5.. (Marks 10) (a) Evaluate the following limit: Lim (θ→0) (1−cos pθ)/(1−cos qθ) (5) (b) Show that dy/dx = y/x if y/x = $\tan^{-1}$ x/y (5)

Q6.. (Marks 10) (a) Evaluate ∫ e^x(1+sin x)/(1+cos x) dx (5) (b) Find an equation of the perpendicular bisector of the segment joining the points A(5,3) and B(9,8) (5)

Q7.. (Marks 10) (a) Evaluate ∫_0^{π} 1/(1+sin x) dx (5) (b) Minimize z = 2x + y, subject to the constraints x + y ≥ 3; 7x + 5y ≤ 35; x ≥ 0; y ≥ 0 (5)

Q8.. (Marks 10) (a) Divide 20 into two parts so that the sum of their squares will be minimum. (5) (b) Find equations of the tangents to the circle $x^2 + y^2 = 2$ parallel to the x−2y+1 = 0 (5)

Q9.. (Marks 10) (a) Find the centre, foci, eccentricity, vertices and directrices of ellipse $x^2 + 16x + 4y^2 - 16y + 76 = 0$ (5) (b) Prove that by vector method sin(α+β) = sin α cos β + cos α sin β (5)